

ECE-GY 6143: Introduction to Machine Learning
Midterm, Fall 2025

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Instructions:

- Exam is closed-book, no calculators or electronic aids.
- You are permitted two double-sided cheat sheets.
- Exam time is 150 minutes
- Write answers on the pages. Write on the front side of the page. Not the back. It will be missed in the scan.
- If you run out of pages, please ask the instructor or course assistant for more pages.

1. *Linear time-series modeling*: You are given scalar time series data, (x_i, y_i) , $i = 0, 1, \dots, T-1$ and want to fit a model:

$$y_i \approx \frac{ax_i}{1 + b_0x_{i-1} + b_1y_{i-1}}, \quad i = 1, \dots, T-1, \quad (1)$$

with parameters (a, b_0, b_1) . Re-arranging (1), we have:

$$y_i(1 + b_0x_{i-1} + b_1y_{i-1}) \approx ax_i. \quad (2)$$

- (a) Using (2), write a linear model of the form:

$$y_i \approx \hat{y}_i = a\phi_{i1}(x, y) + b_0\phi_{i2}(x, y) + b_1\phi_{i3}(x, y), \quad i = 1, \dots, T-1,$$

for three basis functions $\phi_{i1}(x, y)$, $\phi_{i2}(x, y)$, and $\phi_{i3}(x, y)$. Write each $\phi_{ij}(x, y)$, $j = 1, 2, 3$, in terms of the values of x_i , x_{i-1} , y_i and y_{i-1} .

- (b) Given python vectors x and y write python code to create a matrix A with columns of $\phi_{i1}, \phi_{i2}, \phi_{i3}$. Hint: Be careful about the indexing of i vs. $i-1$.

You may use that for a vector $z=(z[0], \dots, z[n-1])$:

$$\begin{aligned} z[a:] &= (z[a], z[a+1], \dots, z[n-1]) \\ z[:b] &= (z[0], z[1], \dots, z[b-1]) \end{aligned}$$

Also,

$$A = \text{np.column_stack}((c0, c1, c2)) \quad \# \text{ matrix with column } c1, c2, c3$$

- (c) Write a python function that, given x and parameters predicts the output $\hat{y}_i$ via the recursion:

$$\hat{y}_i = \frac{ax_i}{1 + b_0x_{i-1} + b_1\hat{y}_{i-1}}, \quad \hat{y}_0 = y_0.$$

You can use the syntax:

```
def predict_rec(x, y0, a, b0, b1):
    ...
    return yhat
```

Inside the function, just loop over i .

$$a) \hat{y}_i = ax_i - b_0x_{i-1}y_i - b_1y_iy_{i-1}$$

$$\phi_{i1}(x, y) = x_i; \quad \phi_{i2}(x, y) = -x_{i-1}y_i; \quad \phi_{i3}(x, y) = -y_iy_{i-1}$$

$$\begin{bmatrix} x_1 & -x_0y_1 & -y_1y_0 \\ \vdots & \vdots & \vdots \\ x_{T-1} & -x_{T-2}y_{T-1} & -y_{T-1}y_{T-2} \end{bmatrix}$$

Continue answer for Problem 1.

```
b) def transform(x,y):  
    n=len(x)  
    col1 = x[1:].reshape(-1,1)  
    col2 = -1 * x[1:n-1] * y[1:].reshape(-1,1)  
    col3 = -1 * y[1:] * y[1:n-1].reshape(-1,1)  
    A = np.column_stack((col1, col2, col3))
```

```
c) def predict_rec(x, y0, a, b0, b1):  
    yhat = np.zeros(len(x))  
    n = len(x)  
    yhat[0] = y0  
    for i in range(1, n):  
        yhat[i] = a * x[i] / (1 + b0 * x[i-1] + b1 * yhat[i-1])  
    return yhat
```


2. *Bias*: Suppose we have a model between scalars x and y :

$$\hat{y} = f(x, \theta) = \theta_0 e^{\theta_1 x}$$

for parameters $\theta = (\theta_0, \theta_1)$.

(a) Given two data points (x_1, y_1) and (x_2, y_2) , find $\hat{\theta}$ such that $y_i = f(x_i, \hat{\theta})$ for $i = 1, 2$.
Hint: Take logarithms and fit a line between the two points.

(b) Now suppose the two data points came from a true function

$$y_i = f_0(x_i) = 1 + e^{x_i}.$$

Note there is no noise. If the data is $x_1 = 0$ and $x_2 = a$ for some $a > 0$, what are $\hat{\theta}_0$ and $\hat{\theta}_1$ in terms of a ?

(c) Given any other $x > 0$, what is the bias error:

$$\text{Bias}(x) = f_0(x) - f(x, \hat{\theta}).$$

Write your answer in terms of x and a .

$$a) \ln(\hat{y}) = \ln(\theta_0) + \theta_1 x$$

$$y_1 = \theta_0 e^{\theta_1 x_1}; y_2 = \theta_0 e^{\theta_1 x_2}$$

$$\downarrow \qquad \qquad \downarrow$$

$$\ln(y_1) = \ln(\theta_0) + \theta_1 x_1 \quad \ln(y_2) = \ln(\theta_0) + \theta_1 x_2$$

$$\ln(y_2) - \ln(y_1) = \theta_1 (x_2 - x_1)$$

$$\boxed{\theta_1 = \frac{\ln(y_2) - \ln(y_1)}{x_2 - x_1}}$$

$$y_2 - y_1 = \theta_0 (e^{\theta_1 x_2} - e^{\theta_1 x_1})$$

$$\boxed{\theta_0 = \frac{y_2 - y_1}{\exp(\theta_1 x_2) - \exp(\theta_1 x_1)}}$$

$$b) \begin{cases} y_1 = 2; x_1 = 0 \\ y_2 = 1 + e^a; x_2 = a \end{cases}$$

$$\theta_1 = \frac{\ln(1 + e^a) - \ln(2)}{a}$$

$$\theta_0 = \frac{e^a - 1}{\exp(\ln(1 + e^a) - \ln(2)) - 1} = \frac{e^a - 1}{\frac{1 + e^a}{2} - 1} = \frac{2e^a - 2}{e^a - 1} = 2$$

Continue answer for Problem 2

$$c) \text{Bias}(x) = 1 + e^x - 2 \exp\left(\frac{\ln(1+e^a) - \ln(2)}{a} x\right)$$

3. *Logistic regression:* Consider a logistic model of the form:

$$z = w_1\phi(x) + w_2\phi(x-3) + b$$

$$\mathbb{P}(y=1|x) = \frac{1}{1+e^{-z}}, \quad \frac{1}{1+e^{-z}}$$

where $\phi(x)$ is the triangular function:

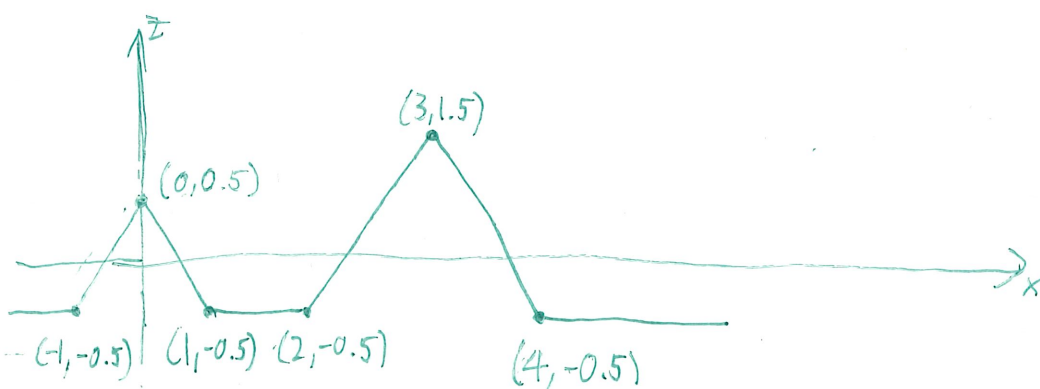
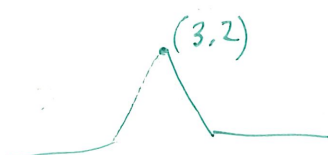
$$\phi(x) = \max\{1 - |x|, 0\}.$$

After training, the parameters are found to be $(w_1, w_2) = (1, 2)$ and $b = -0.5$.

- Draw z as a function for x for the parameters above. The function is piecewise linear. Indicate the precise (x, z) locations of all the line segments.
- Suppose the classifier takes $\hat{y} = 1$ when $\mathbb{P}(y=1|x) > 0.5$. On the graph in part (a), draw the regions of values x where $\hat{y} = 1$. Determine and indicate all the interval boundaries.
- Suppose that for some sample, x is uniformly distributed in the region $x \in [2, 4]$. What is $\mathbb{P}(\hat{y} = 1)$?

$$a) z = \phi(x) + 2\phi(x-3) - 0.5$$

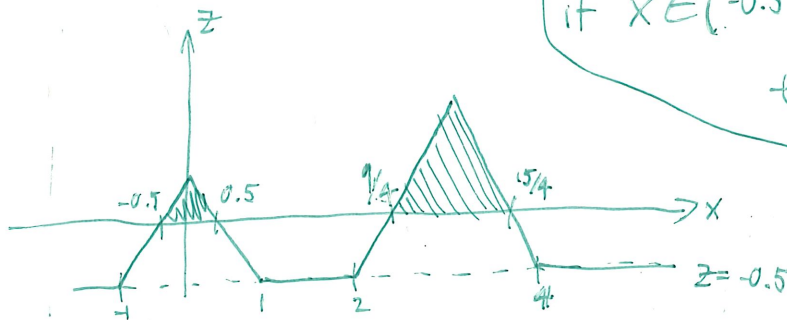
$$= \max(1 - |x|, 0) + \max(2 - 2|x-3|, 0) - 0.5$$



$$2 + \frac{1}{4} = 4 - \frac{1}{4}$$

Continue answer for Problem 3.

b)



if $x \in (-0.5, 0.5) \cup (\frac{9}{4}, \frac{15}{4})$,
then $P(y=1|x) > 0.5$

$$\frac{1}{1+e^{-z}} > 0.5$$

$$1+e^{-z} < 2$$

$$e^{-z} < 1$$

$$-z < 0$$

$$z > 0$$

$$c) x \in [2, 4] \rightarrow \Delta x = 4 - 2 = 2$$

$$\Delta x_{y=1} = \frac{15}{4} - \frac{9}{4} = \frac{6}{4} = \frac{3}{2}$$

$$P(y=1 | x \in [2, 4]) = \frac{3/2}{2} = \frac{3}{4}$$

4. *Gradients*: We wish to fit an inverse model:

$$\hat{y}_i = \frac{1}{1 + \sum w_j x_{ij}}, \quad (3)$$

with a mean squared error metric:

$$J(w) = \sum_{i=1}^n (y_i - \hat{y}_i)^2.$$

- (a) Use chain rule to find the partial derivatives $\partial J / \partial w_j$.
 (b) Complete the function in python that computes the function and its gradient:

```
def Jeval(...):
    ....
    return J, Jgrad
```

You need to specify the arguments for the Jeval.

$$a) \frac{\partial}{\partial w_j} J(\vec{w}) = 2 \sum_{i=1}^n (\hat{y}_i - y_i) \left(\frac{\partial \hat{y}_i}{\partial w_j} \right)$$

$$\frac{\partial \hat{y}_i}{\partial w_j} = \frac{-1}{(1 + \sum_j w_j x_{ij})^2} \cdot (x_{ij}) = \frac{-x_{ij}}{(1 + \sum_j w_j x_{ij})^2}$$

$$\begin{bmatrix} n \times p \end{bmatrix} \begin{bmatrix} p \times 1 \end{bmatrix} = \begin{bmatrix} n \times 1 \end{bmatrix}$$

$$\begin{bmatrix} \frac{\partial y_1}{\partial w_1} \dots \dots \dots \\ \vdots \\ \vdots \end{bmatrix}$$

```
b) def Jeval(x, y, w): # x.shape=(n,p), y.shape=(n,1), w.shape=(p,1)
    dyhdw = -1 * x / ((1 + x.dot(w))←(n,1))**2 # shape (n,p)
    yhat = 1 / (1 + x.dot(w)) # shape (n,1)
    Jgrad = 2 * np.sum((yhat - y)[ :, None] * dyhdw, axis=0) # shape (p,1)
    J = np.sum((y - yhat)**2) # shape (n,1)
    return J, Jgrad
```


Continue answer for Problem 4.

$$Q = \begin{bmatrix} \vec{q}_1 \\ \vec{q}_2 \\ \vdots \\ \vec{q}_m \end{bmatrix}$$

$$Q_{\text{hat}} = \text{np.zeros}((n, p))$$

5. *Nearest neighbor as a transform*: You are given two sets of data samples $x_i \in \mathbb{R}^p$, $i = 1, \dots, n$ and $q_j \in \mathbb{R}^p$, $j = 1, \dots, m$. They are given as python matrices X and Q .

(a) Write python code to create a matrix Q_{hat} with rows:

$$\hat{q}_i = \arg \min_{q \in \{q_1, \dots, q_m\}} \|q - x_i\|,$$

and a vector d_{hat} with values:

$$\hat{d}_i = \|\hat{q}_i - x_i\|^2.$$

That is, for each i , $\hat{q}_i$ is the nearest neighbor in the set $Q \in \{q_1, \dots, q_m\}$ to x_i and d_i is the minimum squared distance. You can use the function:

`np.argmin(d)` % returns i where $d[i]$ is minimum

(b) Now suppose we want to fit a model of the form:

$$\hat{y}_i = w^T \hat{q}_i + \alpha \hat{d}_i,$$

for parameters $w \in \mathbb{R}^p$ and scalar α . So, the prediction of x_i is a linear function of the nearest neighbor $\hat{q}_i$ and distance $\hat{d}_i$. Using the arrays Q_{hat} and d and target vector y , write python code to find parameters w and α . You are given functions:

`beta = np.linalg.lstsq(A, y)` # Finds the LS minima of $\|y - A \cdot \text{dot}(beta)\|$
`A = np.hstack((A1, A2))` # Stacks matrices $A1$ and $A2$ side-by-side

a) $n, p = X.shape$

$m = Q.shape[0]$

$Q_{\text{hat}} = \text{np.zeros}((n, p))$

for i in `range(n)`:

for j in `range(p)`:

$Q_{\text{hat}}[i, j] = \text{np.sum}((Q[j, :] - x[i, :]) ** 2)$

$q_hat_i = \text{np.argmin}(Q_{\text{hat}}, \text{axis}=0)$ # `shape(p,)`

b) $A = \text{np.hstack}((d, Q_{\text{hat}}))$

$\text{beta} = \text{np.linalg.lstsq}(A, y)$

$\alpha = \text{beta}[0]$

$w = \text{beta}[1:]$

$$Q \in \mathbb{R}^{m \times p} \quad \hat{Q} \in \mathbb{R}^{n \times p}$$

$$X \in \mathbb{R}^{n \times p}$$

$$\vec{w}^T \hat{q}_i + \alpha \|\hat{q}_i - \vec{x}_i\|^2$$

$$\begin{bmatrix} d_i & | & Q_{\text{hat}} \end{bmatrix} \begin{bmatrix} \alpha \\ \vec{w} \end{bmatrix} = \begin{bmatrix} y_i \end{bmatrix}$$

Continue answer for Problem 5.

